Example 7.8

The price of a product is related to the number of units available (supply) by the equation $Px+3P-16x=234$, where P is the price of the product per unit in Rupees(₹) and x is the number of units. Find the rate at which the price is changing with respect to time when 90 units are available and the supply is increasing at a rate of 15 units/week.

Solution

We have,

$$P = \frac{234+16x}{x+3}$$

$$\text{Therefore, } \frac{dP}{dt} = -\frac{186}{(x+3)^2}\times\frac{dx}{dt}.$$

Substituting $x=90, \frac{dx}{dt}=15$, we get $\frac{dP}{dt}=-\frac{186}{93^2}\times 15=-\frac{10}{31}\approx -0.32$ rupee/week. That is the price is changing, in fact decreasing at the rate of ₹ 0.32 per week. ■

Example 7.9

Salt is poured from a conveyer belt at a rate of 30 cubic metre per minute forming a conical pile with a circular base whose height and diameter of base are always equal. How fast is the height of the pile increasing when the pile is 10 metre high?

Solution

Let h and r be the height and the base radius. Therefore $h=2r$. Let V be the volume of the salt cone.

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{12}\pi h^3;\ \frac{dV}{dt} = 30\ \text{m}^3/\text{min}.$$

$$\text{Hence, } \frac{dV}{dt} = \frac{1}{4}\pi h^2\frac{dh}{dt}$$

$$\text{Therefore, } \frac{dh}{dt} = 4\frac{dV}{dt}\cdot\frac{1}{\pi h^2}$$

$$\text{That is, } \frac{dh}{dt} = 4\times 30\times\frac{1}{100\pi}$$

$$= \frac{6}{5\pi}\ \text{m/min.}$$

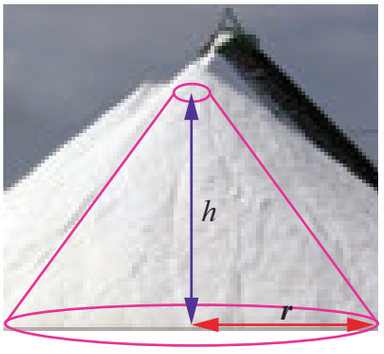


Fig.7.5

■

Example 7.10 (Two variable related rate problem)

A road running north to south crosses a road going east to west at the point P. Car A is driving north along the first road, and car B is driving east along the second road. At a particular time car A is 10 kilometres to the north of P and traveling at 80 km/hr, while car B is 15 kilometres to the east of P and traveling at 100 km/hr. How fast is the distance between the two cars changing?

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